

Optimising Fast Radio Burst Search Pipelines: A Comparative Study of Digital Image Processing Filters for Structural Morphology Preservation

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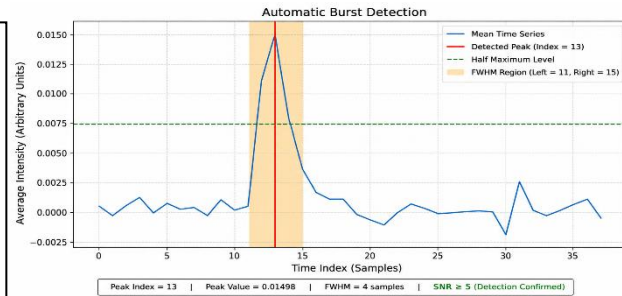
Summary

Led by Dr. Sonam Singh, our group explored the enigmatic cosmic phenomenon of Fast Radio Bursts (FRBs), millisecond-long flashes of radio waves that travel billions of light-years before reaching Earth.

Fast Radio Bursts (FRBs) are mysterious millisecond-long flashes of radio waves that travel billions of light-years before reaching Earth. Hidden within these fleeting signals are clues about some of the Universe's most extreme environments. Yet by the time they reach radio telescopes, they are often buried beneath instrumental noise and terrestrial Radio Frequency Interference (RFI), making reliable detection a major challenge.

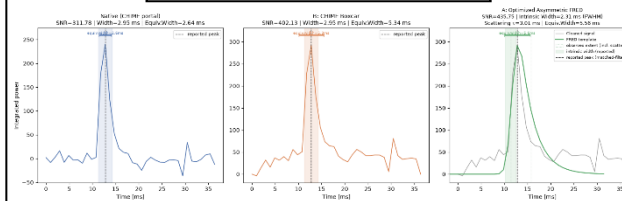
Our work rethinks how FRBs are processed by combining **Digital Image Processing** with radio astronomy. Instead of immediately reducing telescope observations to one-dimensional signals, we preserve their two-dimensional structure through adaptive preprocessing, localized background correction, and morphology-preserving denoising.

We further developed two complementary offline search pipelines, an asymmetric **Exponentially Modified Gaussian (EMG)** model and a conventional **Top-Hat Boxcar** model, to improve burst detection, peak alignment, and signal-to-noise estimation. Benchmarking across **535 real FRBs** from the CHIME/FRB Open Data Catalogue demonstrated consistent improvements over the native CHIME pipeline while providing a scalable and reproducible framework for automated FRB searches.



	Peak SNR	Integrated SNR	Peak Preservation (%)	SSIM	PSNR	Total Wins
Wiener	229	101	196	520	517	1563
Median	151	45	323	0	0	519
Boxcar	46	195	5	0	0	246
Gaussian	76	122	5	0	0	203
Wavelet	28	67	6	15	7	123
Bilateral	5	5	0	0	11	21

Multi-Pipeline Signal Profile -- FRB20180725A



Why this matters

The true value of our pipeline lies in its role as a high-fidelity downstream analysis engine rather than a raw, live streaming filter. Processing 2D images continuously is computationally unfeasible for all-sky real-time backends like CHIME. To resolve this, our framework introduces a tiered strategy: the telescope's native system acts as a fast 1D alarm to flag potential bursts, and our 2D morphology-preserving filters and precise EMG modelling then execute exclusively on those localized data slices. By restricting heavy computational load strictly to confirmed regions of interest, we eliminate wasted processing cycles on empty background noise. This ensures next-generation observatories can maintain maximum streaming speeds while utilizing our pipeline to guarantee that no vital astrophysical data is lost during denoising.